

Literature Status: Almost Complete Coarse Embeddability

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Status: `literature_already_answered`.

Source question

Braga and Chávez-Domínguez ask in Question 4.3 of [1] whether almost complete coarse embeddability is strictly weaker than almost complete $\mathbb{R}$ -isomorphic embeddability. Equivalently, do there exist operator spaces X, Y such that X almost completely coarsely embeds into Y , while X does not almost completely $\mathbb{R}$ -isomorphically embed into Y ?

Identification

Braga, Chávez-Domínguez, and Sinclair explicitly state that their main conceptual result answers this question [2, Introduction and Theorem 1.4]. They construct operator spaces X, Y such that X almost completely isometrically embeds into Y , but X does not even $\mathbb{R}$ -isomorphically embed into Y as a Banach space.

An almost complete isometric embedding is in particular an almost complete coarse embedding, with common control functions $\rho(t) = \omega(t) = t$. Moreover, if there is no real-linear Banach-space embedding $X \rightarrow Y$, then there cannot be a sequence of real-linear maps whose n th amplifications are uniformly isomorphic embeddings. Theorem 1.4 therefore gives an affirmative, strictly stronger answer to the exact source question.

Scope and provenance

The supporting authors knew they were answering Question 4.3 and cite the source paper at the point of the claim. This packet records a literature answer, not a new proof. Their example uses a nonseparable Hilbertian operator space; the source question is existential and imposes no separability condition, so the answer is complete. The supporting paper separately proves a positive linearization statement for separable domains under the stronger almost-complete-isometric hypothesis.

References

- [1] B. M. Braga and J. A. Chávez-Domínguez, *Completely coarse maps are $\mathbb{R}$ -linear*, Proc. Amer. Math. Soc. 149 (2021), 1139–1149; arXiv:2006.00948.
- [2] B. M. Braga, J. A. Chávez-Domínguez, and T. Sinclair, *Lipschitz geometry of operator spaces and Lipschitz-free operator spaces*, Math. Ann. 388 (2024), 1053–1090; arXiv:2104.02714.